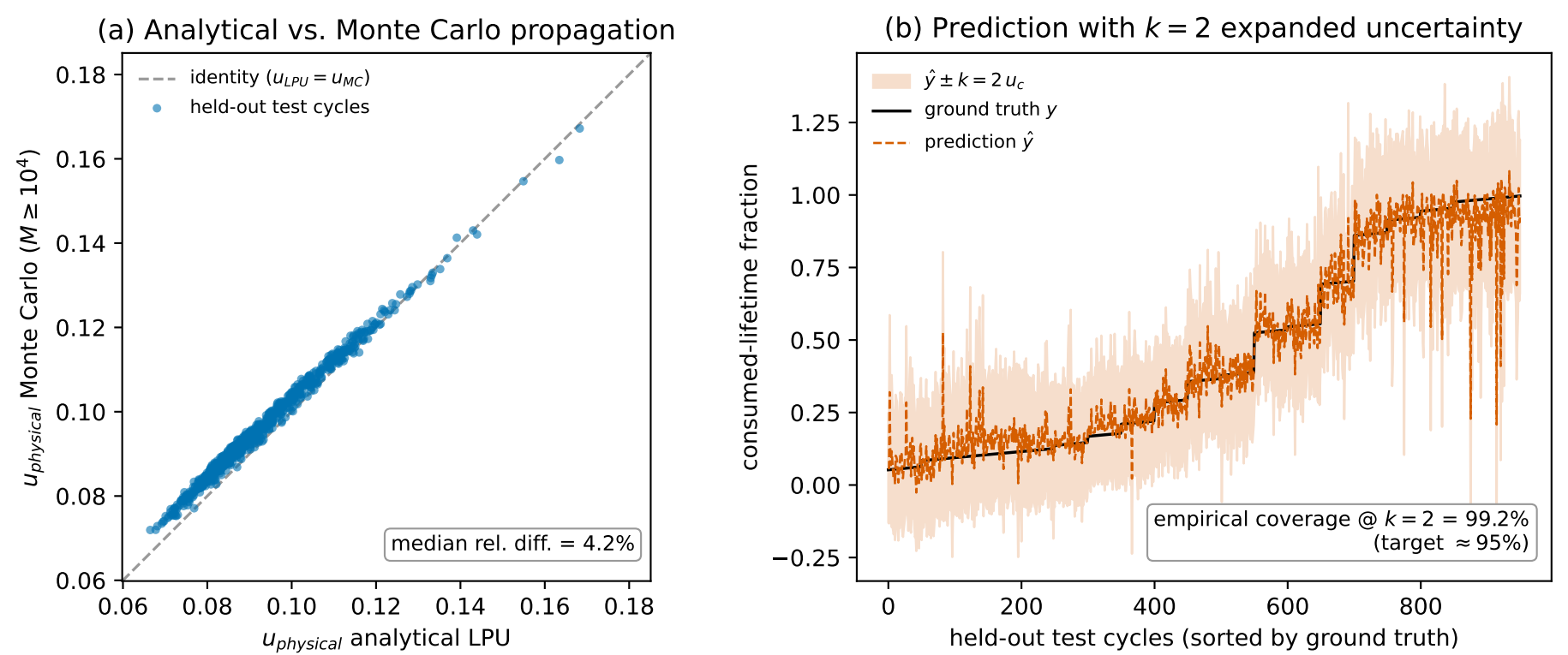

(a) Analytical vs. Monte Carlo propagation
identity ($u_{LPU} = u_{MC}$)
held-out test cycles
median rel. diff. = 4.2%
$u_{physical}$ Monte Carlo ($M \geq 10^4$)
$u_{physical}$ analytical LPU
0.06
0.08
0.10
0.12
0.14
0.16
0.18
(b) Prediction with $k = 2$ expanded uncertainty
$\hat{y} \pm k = 2\,u_c$
ground truth y
prediction $\hat{y}$
empirical coverage @ $k = 2$ = 99.2%
(target ≈ 95%)
consumed-lifetime fraction
held-out test cycles (sorted by ground truth)
−0.25
0.00
0.25
0.50
0.75
1.00
1.25
0
200
400
600
800